

How Does Time Become a Resource?

A Reconstruction of the Concept of Resource through Choice-Possible Time

Chapter 1

Is Time Really a Resource? — Problem Formulation

1.1 The Commodification of Time in Modern Society

In contemporary society, time is widely understood as possessing economic value.

Expressions such as:

- “Time is money”
- time management
- productivity improvement
- efficient use of time
- time as a resource

are commonly used in everyday and economic discourse.

These expressions indicate that time is no longer regarded merely as a natural phenomenon. Rather, it has become an object that can be organized, allocated, evaluated, and optimized within human activities.

In economics, human time has long been treated as an important analytical object. For example, Becker (1965) introduced time into the economic analysis of household behavior by considering both market goods and time as scarce inputs through which individuals maximize utility. His contribution was significant because it expanded economic analysis beyond market transactions and labor supply toward the broader allocation of human time in everyday life.

Similarly, labor economics has continuously examined the allocation of working time, leisure time, and the trade-off between labor and non-labor activities. Human beings are assumed to allocate limited time among various activities, including:

- labor,
- consumption,
- household production,
- leisure,
- and self-development

(Becker, 1965).

However, this conventional understanding contains a fundamental philosophical question:

Is time truly a resource?

Generally, resources are understood as entities that agents can utilize to achieve particular purposes. Economic resources such as land, capital, and raw materials possess several characteristic properties:

- they can be owned,
- they can be stored,
- they can be allocated,
- and they can be accumulated.

Time itself, however, appears fundamentally different.

Time:

- cannot be owned,
- cannot be stored,
- cannot be accumulated,
- and cannot be transferred to others.

For example, people often say that they “have time.” Yet this expression is metaphorical. Unlike land or money, individuals do not literally possess time as an object. Likewise, an unused hour today cannot be stored and consumed tomorrow.

In this sense, time itself does not correspond to the conventional concept of a resource.

Nevertheless, modern society continues to treat time as a resource.

This creates a conceptual tension:

Although time does not appear to satisfy the properties of a resource, society nevertheless uses time as a resource.

This study argues that this contradiction arises because the concepts of “time” and “resource” have not been sufficiently distinguished.

The fundamental question is therefore not:

Is time itself a resource?

Rather, the more appropriate question is:

Under what conditions does time become a resource?

To answer this question, this study reconstructs the resource value of time not by treating time itself as a resource, but by focusing on time intervals that are available to agents for achieving their purposes.

1.2 The Ambiguity of the Concept of Resource

The previous section identified a fundamental problem: although time is treated as a resource in modern society, time itself does not fully correspond to the conventional concept of resources.

To address this problem, it is necessary to clarify what a resource actually means.

However, the concept of resource has never been completely unambiguous.

In classical economics, resources were primarily understood in relation to production activities.

The traditional factors of production were:

- land,
- labor,
- capital.

Smith (1776) analyzed the sources of wealth through division of labor and productive activity, identifying land, labor, and capital as fundamental elements supporting economic production. Ricardo (1817) further examined resource allocation through differences in land fertility and scarcity conditions. Marx (1867) focused on labor power within the production process and analyzed how resources are incorporated into value creation under capitalism.

Within this classical framework, resources were mainly understood as:

inputs that can be introduced into the production process.

Resources referred primarily to concrete entities such as land, raw materials, labor power, and machinery.

However, modern economics has expanded the concept of resource beyond purely material objects.

Resources are increasingly understood as scarce means that individuals and societies can utilize in order to achieve particular objectives.

From this perspective, resources are not limited to physical entities. Information, knowledge, human abilities, institutions, and social relationships may also function as resources when they enable goal achievement.

For example, human capital theory treats skills and knowledge not merely as personal characteristics but as resources generating future productivity and income (Schultz, 1961; Becker, 1964).

Furthermore, Sen's (1999) capability approach emphasizes that the value of resources does not depend solely on the quantity possessed, but rather on what individuals are actually able to achieve through them.

This represents a fundamental transformation of the concept of resource:

From:

Resource = a material object

to:

Resource = an available possibility for achieving purposes

The crucial implication is that the essence of a resource does not reside solely in the existence of an object itself, but in the relationship between the object and an agent.

For example, the value of the same piece of land changes depending on technology, institutions, and purposes of use. Similarly, identical information may have different value depending on the objectives and capabilities of the user.

Therefore, resources should not be understood as objectively fixed objects, but rather as:

relationships through which agents can utilize something to achieve their purposes.

This redefinition allows a new interpretation of time.

Time cannot be owned or stored. However, when an agent possesses a purpose and can utilize a given time interval toward that purpose, that time may function as a resource.

Therefore, the essential question is not:

Is time a resource?

but:

Under what conditions is time transformed into a resource?

The following section distinguishes between time itself and time that functions as a resource.

1.3 Research Question and Problem Setting

The preceding discussion reveals that the relationship between time and resources cannot be resolved through a simple yes-or-no answer.

The fundamental question is:

Is time a resource?

In contemporary society, time is frequently discussed as a resource.

Expressions such as:

- “making effective use of time,”
- “investing time,”
- “saving time”

demonstrate that time is treated as an object possessing economic significance.

However, time itself differs fundamentally from conventional resources.

Resources such as land and capital can be possessed, stored, and allocated for future purposes.

Time, in contrast, continuously passes.

Past time cannot be accumulated, and future time cannot be transferred into the present.

Therefore, regarding time itself as a resource creates conceptual difficulties.

Nevertheless, it would also be incorrect to conclude that time is not a resource.

The reason is that human beings actually use time to achieve purposes.

Humans use time:

- to work,
- to learn,
- to maintain relationships,
- to create,
- and to pursue personal development.

In other words, all human activities exist within temporal conditions.

The crucial distinction is therefore between:

time itself

and

time made available for human purposes.

Time itself is a natural process of temporal passage.

However, when an agent possesses a purpose and can utilize a time interval toward achieving that purpose, that time acquires resource characteristics.

Accordingly, the central research question of this study is not:

Is time a resource?

Rather, it is:

Under what conditions does time become a resource?

To answer this question, this study does not analyze time merely as physical duration.

Instead, the object of analysis is:

time available to an agent for achieving purposes.

This perspective understands time not simply as a flow of physical moments, but as a relationship among:

- agents,
- purposes,
- and possibilities of utilization.

This approach connects with previous discussions in economics.

Becker (1965) incorporated human time allocation into economic analysis by treating time as a scarce constraint in household decision-making. Sen (1999), through the capability approach, emphasized not resources themselves but the possibilities that individuals can realize through them.

These perspectives suggest that the value of resources does not exist inherently within objects but emerges through the relationship between agents and what they are capable of achieving.

This study applies this insight to the concept of time.

The transformation can be represented as follows:

Time

↓

Availability for human utilization

↓

Possibility of achieving purposes

Through this process, time becomes a resource.

The purpose of this study is not to simply add time as a new category of economic resources.

Rather, through examining time, this study seeks to reconsider a more fundamental question:

What is a resource?

1.4 Research Objective and Theoretical Approach

The previous sections have demonstrated that the conventional understanding of time as a resource contains a fundamental conceptual tension. On the one hand, modern societies routinely treat time as a valuable economic resource. Individuals manage time, invest time, save time, and allocate time in pursuit of various objectives. On the other hand, time itself does not possess the essential characteristics traditionally associated with resources. Unlike land, capital, or material goods, time cannot be owned, stored, accumulated, or transferred.

This contradiction suggests that the central issue is not whether time itself should be regarded as a resource, but rather **under what conditions time becomes a resource**.

Accordingly, this study does not attempt to add time to the conventional category of economic resources. Instead, it seeks to reconstruct the concept of resource itself by examining the relationship between time and human agency.

The fundamental argument of this research is that resources are not defined by their physical properties alone. Rather, a resource emerges when a certain entity becomes available for a subject to utilize in order to achieve a purpose. From this perspective, resourcehood is not an inherent property of an object but a relational property established between a subject, a purpose, and a field of possible action.

Applying this framework to time leads to a crucial distinction:

- Time itself is not a resource.
- Time that becomes available for purposeful action can function as a resource.

This distinction allows us to overcome the conceptual difficulty surrounding time. Although individuals cannot possess time itself, they can possess opportunities to act within future temporal intervals. These intervals become meaningful resources when individuals are able to exercise choice regarding how they are used.

Based on this theoretical approach, this study introduces the concept of **Choice-Possible Time**.

Choice-Possible Time refers to the portion of future time during which an individual can select among multiple possible actions according to personal goals and intentions. Unlike conventional measures of time quantity, such as total hours available, Choice-Possible Time emphasizes the qualitative dimension of temporal resources: the degree to which individuals possess meaningful freedom of action within their available time.

This concept provides a new analytical perspective on human time. A person may possess many hours in a day, yet if those hours are completely determined by external obligations, they contain limited resource value. Conversely, even a shorter period of time may possess high resource value if it allows meaningful choice and purposeful action.

Therefore, the value of time does not depend solely on its duration. It depends on the degree of agency embedded within that duration.

1.5 Research Questions

Based on the theoretical problem described above, this study addresses the following research questions.

First:

What conditions are necessary for time to become a resource?

Existing discussions often assume that time is inherently a resource because humans must allocate limited time among competing activities. However, this assumption overlooks the distinction between the passage of time and the availability of time for

intentional action. This study therefore investigates the conditions under which temporal intervals acquire resource value.

Second:

How should the concept of resource be reconstructed in order to explain the resource value of time?

Traditional resource theories have often emphasized scarcity, ownership, and material availability. However, contemporary economic theories have expanded the concept of resources toward information, knowledge, human capability, and institutional arrangements. Building on this development, this study examines whether resources should be understood not as objects possessed by individuals but as possibilities available for achieving purposes.

Third:

How does Choice-Possible Time contribute to understanding human well-being and agency?

Economic systems have traditionally evaluated development through indicators such as production, income, and consumption. However, these indicators do not necessarily capture whether individuals possess meaningful freedom over their own time. This study explores whether Choice-Possible Time can provide an alternative perspective for evaluating human welfare by focusing on the expansion of individual choice possibilities.

Fourth:

What implications does the concept of Choice-Possible Time have in the age of artificial intelligence?

Generative AI technologies have significantly altered the relationship between human activity and time. Rather than creating additional physical time, AI may reduce necessary tasks, accelerate cognitive processes, and expand the portion of time available for human choice. This study examines AI as a technology that transforms the structure of human temporal resources by increasing Choice-Possible Time.

Through these questions, this research aims not merely to redefine time but to reconsider the philosophical foundation of the resource concept itself.

1.6 Structure of the Paper

This paper proceeds as follows.

Chapter 2 examines the historical development of the concept of resources and analyzes how economic theory has traditionally understood resources in relation to production, scarcity, and human activity. It also reviews existing approaches to time in economics, particularly theories of time allocation and human capital.

Chapter 3 critically examines the assumption that time itself is a resource. By comparing the characteristics of conventional resources with those of time, this chapter identifies the conceptual limitations of treating time as a resource without further qualification.

Chapter 4 reconstructs the concept of resource from a relational perspective. It argues that resources are not defined by intrinsic properties but emerge through the relationship between a subject, a purpose, and available possibilities. Based on this framework, the conditions under which time becomes a resource are theoretically derived.

Chapter 5 introduces the concept of Choice-Possible Time and develops its theoretical foundations. This chapter distinguishes between chronological time and resource-generating time, arguing that the resource value of time depends on the degree of choice available to individuals.

Chapter 6 further develops the temporal structure underlying Choice-Possible Time by examining the relationship between future, present, and past. It argues that human life can be understood as the historical allocation and transformation of Choice-Possible Time.

Chapter 7 explores the implications of Choice-Possible Time for contemporary society, particularly in relation to productivity, artificial intelligence, and human well-being. It examines how technological development should be evaluated not merely by increasing output but by expanding human possibilities for meaningful choice.

Finally, Chapter 8 summarizes the theoretical contributions of this study and discusses future research directions. The conclusion argues that time is not inherently a resource. Rather, time becomes a resource when human beings possess the ability to choose and act within it.

The central proposition of this study can therefore be summarized as follows:

Time is not a resource by itself.

Time becomes a resource when human beings can use it to choose their future.

Chapter 2

What Is a Resource?

Reconsidering the Concept of Resource

2.1 Resources as Material Objects

Historically, the concept of resources has been primarily understood in terms of material objects. Particularly in classical economics and resource economics, resources referred mainly to natural and physical entities that supported human production and survival.

Typical examples include:

- oil
- coal
- natural gas
- minerals
- forests
- water resources
- land

These resources exist independently in the natural world and acquire economic value through human extraction, transformation, and utilization.

In classical economic thought, land was regarded as one of the fundamental factors of production alongside labor and capital. For example, Smith (1776) analyzed wealth creation through production and division of labor, while Ricardo (1817) examined the economic significance of land scarcity and differences in productive conditions. In both cases, natural resources were treated as essential foundations of economic activity.

This traditional conception of resources contains several important characteristics.

First, externality.

Resources were understood as entities external to the human subject. They exist independently of whether a particular individual or society chooses to use them. Oil deposits, minerals, and forests exist in nature regardless of human recognition or intention.

Second, ownership.

Traditional resources possess the characteristic of ownership. Land, minerals, and other natural resources can be legally controlled through property rights. The question of who owns and manages resources has therefore been a central issue in economic theory and institutional analysis (Demsetz, 1967).

Third, storability.

Many material resources can be stored and preserved. Oil can be stored in tanks, minerals can be kept in warehouses, and unused resources can be carried forward into the future. Their availability can therefore be maintained beyond the moment of acquisition.

These three characteristics have historically formed the foundation of the conventional concept of resources.

Characteristic	Meaning
Externality	Exists independently of the subject
Ownership	Can be possessed or controlled institutionally
Storability	Can be preserved and carried into the future

Under this traditional framework, resources are primarily understood as **objects that exist in the world and can be utilized by humans.**

However, applying this definition directly to time creates a conceptual difficulty.

Although time exists independently as a natural phenomenon, it cannot be owned, accumulated, or stored. Unused time cannot be transferred into the future, nor can temporal availability be preserved in the same way as material resources.

Therefore, a resource concept based exclusively on material properties cannot adequately explain the common economic understanding that "time is a resource."

This problem does not merely reveal the uniqueness of time. Rather, it indicates a limitation within the traditional concept of resources itself: the concept was historically constructed around material objects.

Therefore, the following section examines resources from a broader perspective—not as physical objects, but as forms of **availability for purposeful human action**.

2.2 Resources in Economic Theory

The previous section examined the historical understanding of resources as material entities. However, modern economics has gradually expanded the concept of resources beyond land and physical materials.

Particularly since the twentieth century, economic theory has increasingly incorporated non-material factors such as time, knowledge, and human capabilities into its analysis of resource allocation.

A landmark contribution in this development is Becker's (1965) theory of the allocation of time.

Becker argued that human utility is not generated solely through market goods but through activities that combine goods and time. For example, eating requires not only food but also time spent preparing and consuming meals. Similarly, education, childcare, household production, and leisure activities all require temporal inputs.

Within this framework, time is treated as a fundamental constraint alongside income. Individuals are understood as allocating their limited daily time among labor, leisure, household activities, and self-development in order to maximize utility (Becker, 1965).

This approach was theoretically significant because it introduced time as a central element of economic analysis.

Furthermore, human capital theory expanded this perspective by interpreting time invested in education and training as an investment that generates future economic returns (Becker, 1964). Human time became understood as an important factor contributing to value creation.

Thus, modern economics recognizes time as an economically significant resource supporting production, consumption, and human capital formation.

However, an important conceptual issue remains unresolved.

Becker's theory explains **how individuals allocate time among competing activities**. In this framework, time functions as an input into economic activities.

Yet, the following two statements are not necessarily equivalent:

Time is used as an input in economic activities.

and

Time itself is a resource.

For example, electricity used in a factory is an input into production. However, the fact that electricity is consumed in production does not by itself explain why electricity constitutes a resource. Similarly, the fact that time is allocated to production or consumption does not fully explain the conceptual basis for considering time a resource.

In other words, Becker's theory successfully explains the economic allocation of time, but it does not directly answer the deeper question:

Under what conditions does time become a resource?

This question lies at the center of the present study.

The fact that humans utilize time in economic activities is undeniable. However, the resource value of time may not originate from the intrinsic properties of time itself. Rather, it may emerge from the relationship between human subjects, their purposes, and their ability to utilize available time.

Therefore, this study accepts time allocation theory as a crucial foundation while reconsidering the underlying concept of resource itself.

The fundamental question is not:

How should time be allocated?

but rather:

Under what conditions does time become a resource?

2.3 The Essence of Resources: Availability for Purposeful Action

The previous discussion has shown that resources have historically been understood as material objects, while modern economics has expanded the concept to include non-material entities such as time, knowledge, and human capabilities.

However, this expansion raises a fundamental question:

Why can such fundamentally different entities as oil and time both be described as "resources"?

To answer this question, the essence of resources must be reconsidered beyond material properties.

This study defines a resource as follows:

A resource is a possibility available to a subject for achieving a purpose.

This definition emphasizes that resources are not objects possessing value independently. Rather, resources emerge through the relationship between a subject and a purpose.

Oil becomes a resource not because it exists physically, but because it can be utilized as fuel or as a material for industrial production. Knowledge becomes a resource when it contributes to solving problems. Human abilities become resources when they can be applied toward meaningful objectives.

Therefore, the existence of a resource requires at least three conditions.

(1) Subject

The first condition is the existence of a subject capable of utilization.

Without a subject, utilization cannot occur. Oil deposits and land may exist in nature, but without a subject capable of employing them, they remain merely natural entities.

A resource emerges only through its relationship with a subject.

(2) Purpose

The second condition is the existence of a purpose.

Resources always acquire meaning in relation to objectives. The same piece of land may function as agricultural land for farmers, residential space for urban planners, or a conservation target for ecologists.

Therefore, resource value is not an objective property independent of human purposes. It depends on what the subject seeks to achieve.

(3) Availability

The third condition is practical availability.

An object cannot function as a resource unless it is actually available for utilization. Oil located deep underground may exist, but without extraction technology it cannot serve as a usable resource. Knowledge that cannot be understood cannot contribute to action. Human abilities that cannot be exercised under given conditions cannot function as resources.

Thus, existence and availability are fundamentally different concepts.

Availability refers to the condition in which a subject can select and employ something for the achievement of a purpose.

These three conditions can be summarized as follows:

Necessary condition	Meaning
Subject	A user capable of utilization exists

Necessary condition

Meaning

Purpose

A goal to be achieved exists

Availability

The object can be applied toward that goal

None of these conditions can be omitted.

Availability without a subject does not constitute a resource. A subject without a purpose cannot determine what should be considered valuable. A purpose without availability cannot be realized through actual utilization.

The theoretical contribution of this study is to redefine resources not as properties inherent in objects, but as **relational phenomena emerging from the interaction between subject, purpose, and availability**.

This reconstruction enables a unified analysis of material resources, knowledge, capabilities, information, and ultimately time.

The implication for time is fundamental.

Time cannot be owned or stored. Nevertheless, when a subject possesses a purpose and can meaningfully utilize a temporal interval, that time can function as a resource.

Thus, the question is not whether time itself is a resource.

The essential question is:

Under what conditions does time become a resource?

Proposition 1

A resource is not an inherent property of an object.

It emerges through the relationship in which a subject can utilize something for the achievement of a purpose.

Chapter 3

Why Time Itself Does Not Become a Resource

From Flowing Time to Resource Intervals

3.1 Flowing Time and the Concept of Resource

The previous chapter demonstrated that a resource should not be understood as an intrinsic property of an object, but rather as a possibility that becomes available to a subject for achieving a purpose. Based on this definition, whether time can function as a resource cannot be determined by examining the nature of time itself. Instead, it requires an analysis of the relationship between human beings and time.

For this purpose, it is first necessary to clarify the philosophical understanding of time.

Time has long been a central subject of philosophical inquiry. However, many philosophers have not regarded time as an independent object existing in the world in the same manner as physical entities.

Aristotle, in *Physics*, defined time as “the number of motion in respect of the before and after” (Aristotle, trans. 1984). According to this view, time is not a substance existing independently from change and motion. Rather, time emerges through the measurement and ordering of movement. Without change or motion, time would not become intelligible as an object of human understanding.

In contrast, Heidegger approached time from an existential perspective. In *Being and Time*, he argued that time is not merely a measurable sequence of moments but the fundamental structure of human existence itself, which he called **temporality** (Heidegger, 1927/1962). Human beings exist through their orientation toward future possibilities, their relationship with the past, and their engagement with the present. Temporality is therefore not something external to human existence but the very structure through which human beings understand themselves and the world.

Although Aristotle and Heidegger developed fundamentally different theories of time, they share an important insight: time is not a self-standing object that simply exists independently in the world.

For Aristotle, time exists through its relation to movement.

For Heidegger, time exists through the structure of human existence.

In both cases, time is understood relationally rather than as an independent entity.

This philosophical insight has significant implications for the concept of time as a resource. Resources are generally understood as objects that can be utilized, allocated, managed, or exchanged. However, time itself does not possess these characteristics. Time continuously passes; it cannot be stopped, stored, accumulated, or owned.

Human beings do not control the flow of time itself. Rather, they exist within the flow of time.

Therefore, the proposition that “time is a resource” cannot be accepted without further theoretical clarification.

However, this study does not attempt to answer the ontological question of “what is time?” Nor does it deny the obvious fact that time passes. The central question of this research is different:

Among the continuous flow of time, under what conditions does a portion of time become available for human purposes and function as a resource?

In other words, a theoretical gap exists between philosophical theories of temporality and economic theories that treat time as a resource. Philosophy has explained the nature of time and human existence, while economics has analyzed time allocation and constraints. Yet the conditions under which time itself becomes a resource have remained insufficiently examined.

This chapter addresses this missing link.

3.2 The Non-Ownership of Time

The previous section established that time is not a physical entity but a phenomenon that emerges through the relationship between human beings and the world. This section examines one of the most important consequences of this characteristic: the non-ownership of time.

Traditionally, resources have been understood as entities that can be possessed, controlled, and utilized by subjects according to their purposes. Land can be owned through property rights. Money can be stored, transferred, and accumulated. Many physical resources share the common characteristic that they can be retained and managed for future use.

Time, however, fundamentally differs from such resources.

People commonly use expressions such as:

- “my time”
- “free time”
- “I lost time”

Yet these expressions do not indicate actual ownership of time.

Unlike land or money, time cannot be possessed as an object. A person does not own time itself; rather, every person experiences the irreversible passage of time.

For example, unused land can remain available for future use. Unused money can be saved and transferred. However, an unused hour cannot be stored and recovered tomorrow. A moment that has passed cannot be retrieved or transferred to another person.

Time is therefore fundamentally non-transferable and non-accumulative.

This understanding is consistent with philosophical theories of time. Heidegger conceptualized time as the temporal structure of human existence rather than an external object possessed by individuals (Heidegger, 1927/1962). Similarly, Bergson criticized the treatment of time as a spatialized object and emphasized lived duration (*durée*) as the subjective experience of continuous becoming (Bergson, 1910/2001).

Economic theories also implicitly recognize this distinction. Becker’s time allocation theory treats time as a finite constraint faced by individuals, not as a commodity that can be owned like capital or land (Becker, 1965).

Therefore, the everyday statement that “time is a resource” cannot be accepted literally.

Time itself cannot be owned, stored, or transferred.

However, this does not mean that time can never function as a resource.

Rather, the fundamental question is:

Under what conditions can non-ownable time acquire resource characteristics?

This study argues that the answer cannot be found in time itself, but in the relationship between human purposes and the availability of time for action.

3.3 The Transformation into Time Intervals

The previous section demonstrated that time itself cannot be owned or accumulated and therefore differs from conventional resources. However, human beings clearly utilize time in everyday life.

How can this apparent contradiction be explained?

This study argues that the key lies in the human practice of dividing continuous time into meaningful intervals.

Natural time flows continuously. The universe itself does not contain categories such as “9 a.m.,” “working hours,” or “holidays.” These divisions are not properties of time itself but social constructions created to organize human activities.

Examples include:

- working hours from 9 a.m. to 5 p.m.
- school class periods
- deadlines
- scheduled meetings
- medical appointments
- vacations and holidays

All of these represent the transformation of continuous temporal flow into bounded units.

At this moment, the object utilized by humans is no longer “time itself.” Rather, it becomes a defined temporal segment.

This study refers to such segments as:

Resource Intervals

A **Resource Interval** is defined as:

A socially and institutionally defined temporal segment that a subject separates from continuous time and treats as an available unit for achieving a particular purpose.

The introduction of this concept clarifies the relationship between time theory and resource theory.

Time can therefore be distinguished into two levels:

Concept	Characteristics
Flowing Time	Continuous temporal passage without boundaries
Resource Interval	A bounded temporal unit created for human purposes

This distinction is crucial.

The common statement “time is a resource” does not actually refer to flowing time itself. What humans manage, schedule, contract, and allocate are not abstract time but socially defined temporal intervals.

For example, an employment contract does not exchange “time” in general. It specifies a particular interval, such as working from 9 a.m. to 5 p.m. An airline ticket does not purchase “three hours of time” in the abstract; it provides access to a specific interval between departure and arrival. Education, healthcare, transportation, and administrative systems are all organized around structured temporal intervals.

Therefore, human society does not directly utilize time itself. Instead, society transforms continuous time into manageable units through the process of temporal segmentation.

However, this study does not define the resource interval itself as a complete resource.

Creating a time interval is a necessary condition for resource formation, but it is not sufficient.

Consider three hours spent on an airplane. This is certainly a temporal interval. Yet if the individual has almost no ability to determine how that time is used, its resource value is limited.

Similarly, compulsory waiting time or highly constrained working time may constitute temporal intervals, but they do not necessarily represent valuable resources for the individual.

Thus:

A resource interval is the first stage of transforming time into a resource.

A further condition is required for a temporal interval to become a genuine human resource.

This study identifies that condition as **choice possibility**.

The next chapter introduces the concept of **Choice-Possible Time**, which explains when a temporal interval becomes a resource through the existence of meaningful alternatives and individual agency.

Chapter 4

How Do Time Intervals Become Resources?

The Condition of Choice Possibility

4.1 A Time Interval Alone Does Not Constitute a Resource

The previous chapter introduced the concept of the **Resource Interval**, demonstrating that human societies transform continuous temporal flow into bounded units that can be socially organized and utilized. However, the mere existence of a time interval does not necessarily mean that time has become a resource.

Consider a possible objection to the argument developed so far.

If being a time interval were sufficient for becoming a resource, then all time intervals should possess equivalent resource value. Yet this clearly contradicts ordinary human experience.

For example:

- three hours spent on an airplane,

- two hours spent in a mandatory meeting,
- one hour waiting in a hospital,
- time spent under imprisonment,

are all clearly identifiable temporal intervals. Nevertheless, most people would not regard these periods as freely available resources.

Of course, individuals may engage in certain activities during these periods. A passenger may read a book during a flight, or a participant may reflect on ideas during a meeting. However, the range of possible actions is significantly constrained. Individuals cannot freely design these periods according to their own purposes, nor can they flexibly allocate them among alternative activities.

This demonstrates that the concept of a Resource Interval alone cannot fully explain the resource value of time.

The present study argues that the missing condition is **choice possibility**.

As defined in Chapter 2, a resource is:

A possibility available to a subject for achieving a purpose.

Accordingly, “availability” does not simply mean that a period of time exists. It means that a subject possesses the ability to choose among multiple possible actions within that period.

Consider two periods of equal duration: three hours spent on an airplane and three hours spent freely at home.

Physically, they are identical in length.

However, their resource values differ substantially because the range of possible actions available to the individual differs. During a flight, physical and institutional constraints limit possible activities. In contrast, at home, individuals may choose among various possibilities, such as working, studying, resting, engaging in hobbies, or spending time with family.

Therefore, the value of time as a resource is not determined by its quantity but by the degree of choice available within that time interval.

This perspective shares a certain theoretical affinity with Amartya Sen’s capability approach. Sen argued that human well-being should not be evaluated merely by the

possession of goods or income but by the substantive freedoms individuals possess: what they are actually able to do and to be (Sen, 1999).

Similarly, this study argues that the value of time should not be measured by duration alone but by the extent to which a given period provides meaningful possibilities for action.

However, this study does not simply apply the capability approach to time.

The central claim is different:

Choice possibility is not merely a criterion for evaluating resources; it is a necessary condition for something to become a resource in the first place.

Therefore, this study derives the following proposition:

A Resource Interval is a necessary condition for time to become a resource, but it is not sufficient. A time interval becomes a resource only when a subject possesses the possibility to choose actions according to their purposes.

This proposition leads directly to the central concept of this study: **Choice-Possible Time**.

The following section defines this concept and identifies the sufficient conditions under which time becomes a resource.

4.2 Choice Possibility as the Condition of Resource Formation

The previous section demonstrated that the formation of a Resource Interval does not automatically create a resource. A flight duration or mandatory meeting period may constitute a clearly defined interval, yet its resource value remains limited because individuals cannot freely determine how that time is used.

This indicates that the resource value of time depends not on the existence of a temporal interval itself, but on the range of actions available to the subject within that interval.

Therefore, this study introduces **choice possibility** as the fundamental condition through which time becomes a resource.

The concept of **Choice-Possible Time** is defined as follows:

Choice-Possible Time refers to a period during which a subject can select and perform actions among multiple possible alternatives according to their own purposes.

This definition contains three essential components.

First, the existence of a subject

Choice requires an agent capable of making decisions. Without a subject, there can be no meaningful selection among alternatives. A period of time without an acting subject cannot constitute a resource.

Second, the existence of a purpose

Choice is always related to goals. Without some purpose or desired state, alternative actions have no significance. A temporal interval becomes meaningful as a resource only when it is connected to what the subject seeks to achieve.

Third, the existence of multiple actionable possibilities

The subject must actually possess alternative possibilities within that time interval. When physical, institutional, or social constraints eliminate meaningful alternatives, the resource value of that time declines.

According to this definition, even identical durations may possess radically different resource values.

For example, three hours during which an individual may freely choose between reading, researching, resting, or interacting with family represent a high level of Choice-Possible Time.

Conversely, three hours during compulsory transportation or mandatory participation may constitute a temporal interval but contain limited choice possibility.

Therefore, this study proposes the following proposition:

A Resource Interval is necessary for time to become a resource, but sufficient resource formation requires the existence of choice possibility.

This perspective differs from conventional economic approaches that primarily treat time as a quantitative input.

In economic analysis, time is often modeled as a limited quantity allocated among competing activities. However, this study argues that the resource value of time is not determined by duration alone. Rather, it is determined by the qualitative structure of possible actions available within that duration.

Thus:

The value of time does not lie in the amount of time possessed, but in the possibilities that can be realized within that time.

The concept of Choice-Possible Time introduced here becomes the foundation of the theoretical model developed in the next section.

4.3 The Time Resource Formation Model

The previous sections have demonstrated that time itself is not a resource. Furthermore, transforming continuous time into bounded intervals does not automatically create a resource.

This study argues that time becomes a resource only when several conditions are satisfied sequentially.

The first condition is the transformation of continuous temporal flow into a **Resource Interval**.

Human societies divide continuous time through institutions, contracts, and everyday practices. Working hours, holidays, appointments, and deadlines are examples of socially constructed temporal intervals.

However, a Resource Interval is only a prerequisite for resource formation. It does not itself constitute a resource.

The second condition is the existence of a subject who can utilize the interval.

Resources do not exist independently from subjects. A temporal interval acquires meaning only through an individual or collective actor who seeks to use it.

Furthermore, the subject must possess a purpose.

Only when a purpose exists can a temporal interval represent a possibility for achieving something.

The most important condition, however, is the existence of choice possibility.

A subject must be able to select among multiple possible actions within the given interval. When actions are predetermined and no meaningful alternatives exist, the time interval has little resource value.

Integrating these arguments, this study proposes the following model:

Continuous Time



Temporal Segmentation



Resource Interval



Subject



Purpose



Choice Possibility



Resource Formation

This model demonstrates that the transformation from time into resource is not a simple process of:

Time $\rightarrow$ Time Resource

Rather, time becomes a resource only through a sequence of relational conditions:

1. continuous time is socially transformed into intervals;
2. a subject exists who can utilize those intervals;
3. the subject possesses purposes;
4. meaningful choices among alternative actions are available.

Therefore, the resource character of time is not an intrinsic property of time itself. It is a relational property emerging from the interaction between subject, purpose, and choice possibility.

This argument also generalizes the broader theory of resources proposed in Chapter 2.

Resources are not substances possessing inherent value. They are relational possibilities created when a subject can utilize something for achieving a purpose.

Time follows the same principle.

Time becomes a resource only when the relationship between subject, purpose, and choice possibility is established.

The next chapter further develops this argument by analyzing Choice-Possible Time as a measure of the value of time resources and examining how the expansion of such time can become a fundamental indicator of human well-being.

Chapter 5

Reconstructing the Concept of Time through Choice-Possible Time

5.1 What Is Choice-Possible Time?

The previous chapter demonstrated that time itself does not constitute a resource. Rather, time becomes a resource only when a time interval satisfies three conditions: the existence of a subject, the presence of a goal, and the availability of meaningful choices within that interval.

This chapter introduces the central concept of this study: **Choice-Possible Time**.

In this study, Choice-Possible Time is defined as:

Choice-Possible Time refers to a period in which a subject can choose among multiple possible actions in accordance with personal goals and can realize those choices in the future.

The essential point of this definition is that the value of time does not depend primarily on its quantity, but on the degree of choice available within that time.

For example, three hours may have completely different resource values depending on the conditions under which they occur. Three hours spent freely reading, conducting research, resting, or engaging with family members provide a wide range of possible actions. Such time possesses a high degree of choice availability.

In contrast, three hours spent on an airplane, in a compulsory meeting, or under institutional confinement may constitute a clear time interval, but the range of possible actions available to the individual is significantly restricted. Although the person physically experiences the same duration of time, the resource value of that time is considerably lower.

Therefore, the relevant object for evaluating time as a resource is not merely the amount of time available. Rather, it is the extent to which a person maintains an open space of possible choices toward the future.

In this respect, Choice-Possible Time differs fundamentally from the conventional concept of **free time**. Free time is usually defined negatively as time released from compulsory activities such as labor or household duties. However, the present study argues that free time does not necessarily correspond to genuine resource value.

A person may formally possess free time but lack meaningful choices because of exhaustion, financial constraints, social obligations, or psychological limitations. Conversely, even within a constrained situation, time may possess high resource value if an individual can make meaningful decisions according to personal purposes.

Thus, Choice-Possible Time is not an intrinsic property of time itself. It is a relational concept emerging from the interaction among:

- the subject,
- the subject's goals,
- and the available possibilities for action.

Although time flows uniformly as a physical phenomenon, its value as a resource differs according to the size of the choice space that remains open to the individual.

Furthermore, Choice-Possible Time is fundamentally a future-oriented concept.

Past time has already been consumed, while present time represents the moment in which choices are being executed. Future time, however, contains unrealized possibilities among which choices can still be made.

Therefore, in this study, time as a resource is understood as:

the totality of future-oriented possibilities available for meaningful choice.

The following sections further examine this temporal structure and demonstrate that the essence of time as a resource lies not in the passage of time itself, but in the possibilities that remain open toward the future.

5.2 Distinction from Free Time

The previous section defined Choice-Possible Time as a period during which a subject can select and realize possible actions according to personal goals. This section distinguishes this concept from the conventional notion of **free time**.

Free time is generally understood as time free from obligatory activities such as employment or household responsibilities. In economics and leisure studies, it has often been treated as the counterpart of working time, and an increase in leisure time has been regarded as an indicator of improved welfare (Becker, 1965; Gershuny, 2000).

However, this definition does not sufficiently distinguish between being free from obligations and possessing genuine freedom of action.

This study argues that these two concepts must be separated.

For example, an individual may obtain three hours of free time after work but, due to extreme fatigue, may realistically have no option other than rest. Similarly, economic

constraints may prevent a person from choosing education, travel, cultural activities, or other personally meaningful pursuits.

The same issue appears in situations involving childcare, eldercare, or other forms of responsibility. Although individuals may technically possess free time, their actual range of possible actions may remain highly restricted.

Such time is categorized statistically as free time, yet its value as a resource is limited because the individual's effective choice space is narrow.

Conversely, time subject to certain constraints may possess substantial resource value if individuals retain meaningful decision-making capacity and can act according to their purposes.

Therefore, the crucial question is not whether time is free from obligations, but whether the time provides an individual with meaningful possibilities for action.

Accordingly, this study distinguishes:

$$\text{Free Time} \neq \text{Choice-Possible Time} \quad \text{\textbf{Free Time}} \neq \text{\textbf{Choice-Possible Time}}$$

Free Time represents a social or institutional classification of time, whereas Choice-Possible Time represents the actual availability of meaningful choices within that time.

The former concerns the external condition of time; the latter concerns the relational condition between subject, purpose, and available possibilities.

This distinction has important theoretical implications.

If time resources are understood merely as free time, increasing the quantity of leisure hours would automatically imply greater well-being. However, actual human experience demonstrates that additional free hours do not necessarily increase human capability when individuals remain constrained by exhaustion, poverty, or social limitations.

Therefore, the relevant measure is not the quantity of free time but the amount of Choice-Possible Time that individuals possess toward their future.

Only through this perspective can the resource value of time be explained in terms of actual human agency.

5.3 A Model of Time Resource Value

The previous sections demonstrated that the value of time as a resource cannot be determined solely by the quantity of time available.

Traditional approaches to time allocation have primarily focused on how individuals distribute limited hours among competing activities. However, this study argues that identical amounts of time can possess significantly different resource values depending on the conditions surrounding the individual.

To conceptualize this difference, this study proposes the **Time Resource Value Model**.

Conceptually, the value of time as a resource is determined by three factors:

$$\text{Time Resource Value} = \text{Time Quantity} \times \text{Choice Possibility} \times \text{Goal Relevance}$$

Each component is defined as follows:

Time Quantity

The length of the time interval available to the subject.

Choice Possibility

The degree to which the subject can select among multiple possible actions.

Goal Relevance

The extent to which the available time contributes to achieving the subject's purposes.

The central implication of this model is that time quantity alone cannot explain the value of time resources.

For example, ten hours spent under strict obligation may provide little resource value if individual actions are predetermined and decision-making possibilities are minimal.

Conversely, three hours devoted to research, creativity, learning, or meaningful interaction may possess substantially higher resource value because the individual can actively employ the time toward personally significant goals.

Conceptually:

Ten hours of constrained time

Limited choice possibility

↓

Low goal relevance

↓

Low time resource value

Three hours of creative activity

High choice possibility

↓

High goal relevance

↓

High time resource value

Therefore, time should not be understood as a homogeneous physical quantity. Its resource value emerges through the relationship between temporal availability and human agency.

The Time Resource Value Model does not measure time merely by duration. Instead, it evaluates the extent to which individuals can realize meaningful possibilities toward the future.

It should be noted that the equation presented here is a conceptual model rather than an empirically validated measurement formula. Future research must examine how choice possibility and goal relevance can be operationalized and quantified.

Nevertheless, the model provides a theoretical framework that extends beyond traditional approaches focusing solely on the amount of available time. It integrates time, human agency, and purpose into a unified understanding of resources.

Ultimately, this model supports the central argument of this study:

Time itself is not a resource. Time becomes a resource when a subject can exercise meaningful choices toward a purpose, and the value of that resource is determined by the degree of choice and relevance embedded within that time.

Chapter 6

A Temporal Theory of Resources: Future, Present, and Past Time

6.1 Future Time

This study has argued that time becomes a resource only when three conditions are satisfied: the existence of a subject, the presence of a goal, and the availability of meaningful choices. From this perspective, the conventional distinction between past, present, and future should not be understood merely as a chronological sequence. Rather, each temporal dimension possesses a different relationship with resource potential.

First, future time is not simply time that has not yet arrived. In this study, future time is understood as a domain in which a subject retains multiple possible courses of action.

In other words, the future is a space of open possibilities.

Consider an upcoming holiday next week. At that moment, an individual may choose to travel, read a book, conduct research, or spend time with family. These possibilities coexist because no final decision has yet been made. The essential characteristic of future time lies precisely in this openness and indeterminacy.

This understanding directly corresponds to the concept of **Choice-Possible Time** proposed in this study. As long as a subject maintains multiple possible choices toward the future, that time possesses the potential to function as a resource.

Conversely, if future time is completely predetermined and the individual has virtually no ability to make meaningful choices, the existence of that time interval alone does not guarantee resource value.

Therefore, this study conceptualizes future time not merely as "time that has not yet occurred," but as:

a potential resource space in which unrealized possibilities remain available to the subject.

Future time itself is not yet a resource. However, when a subject retains the ability to choose and act according to personal goals, future time can become a resource.

From this perspective, the future should not be regarded simply as a portion of time that lies ahead. Rather, it should be understood as a space of possibilities through which individuals construct their own actions and identities.

Future time is not merely unused time. It is a field of potential choices that may become resources through human agency.

6.2 Present Time

The previous section defined future time as a domain in which possibilities remain open. This section examines the nature of present time.

The present is often understood simply as "the current moment." However, such a definition describes only the temporal position of an event and does not sufficiently explain its relationship with human action.

This study defines the present as:

the process through which choices are actualized into action.

Although multiple possibilities exist in the future, they become concrete realities only through decisions and actions occurring in the present.

At the present moment, a subject selects one possibility among those previously available and transforms it into an actual activity.

For example, on a holiday morning, an individual may have several possible choices: reading, going outside, conducting research, or spending time with others. However, once the individual begins writing a research paper at a desk, one possibility has been realized, while the others remain unrealized.

Thus, the present is not merely a point located between the past and future. It is a dynamic process in which possibilities are transformed into reality.

From this perspective, present time serves as the mediating stage connecting future possibilities and past outcomes.

Possibilities that existed in the future are realized through present decisions and actions, and these realized actions subsequently become part of the past.

Therefore, this study proposes a three-layer temporal structure of time resources:

- Future time: time as potential choice
- Present time: time as the execution of choice
- Past time: time as the accumulated record of realized choices

Within this structure, present time represents the practical domain where individuals convert temporal possibilities into actual value.

Accordingly, present time in this study does not refer simply to "the current instant." It refers to:

the process through which subjects transform future possibilities into meaningful actions.

6.3 Past Time

The previous section positioned future time as open possibility and present time as the process of realizing choices. This section examines past time from the perspective of resource theory.

Past time is commonly understood as time that has already elapsed. However, this study argues that such an interpretation is insufficient.

Past time should be understood as:

the accumulated history of choices that have already been realized and whose possibilities have been consumed.

Future time contains multiple possible pathways. Through decisions and actions in the present, however, these possibilities become fixed as actual events.

At this moment, alternative possibilities that were not selected disappear, while the chosen action remains as part of personal history.

Therefore, the past is not merely a collection of completed events. It is the historical record of consumed Choice-Possible Time.

From this perspective, human life is not simply the passage of chronological time.

Rather:

Life is the history of how individuals have allocated and utilized their limited Choice-Possible Time.

This definition is consistent with the temporal resource model proposed in this study.

Time exists as possibility in the future, becomes action in the present, and becomes irreversible history in the past. Time resources are not merely consumed; the manner in which they are consumed constitutes the formation of the individual's life trajectory.

Furthermore, this perspective is compatible with the view that the subject is not a fixed entity but a continuously generated process.

Individuals construct themselves by allocating time according to their values and goals. The past therefore functions not merely as a record of events but as evidence of how individuals have historically used their temporal resources.

Accordingly, this study defines past time not as "time that has ended," but as:

the accumulated history of Choice-Possible Time that has been transformed into experience.

In this sense, human life is not measured by the amount of time one possesses, but by how one has transformed available time into meaningful choices.

The past is no longer a resource itself.

However, it becomes an experiential resource through which individuals revise their future values, goals, and choices.

Chapter 7

Time Resources in the Age of Artificial Intelligence

7.1 Does AI Increase Human Time?

The recent development of generative artificial intelligence (AI) has brought significant transformations to human intellectual activities. Tasks such as writing, information retrieval, programming, translation, and even complex reasoning assistance can now be performed in substantially shorter periods of time. Consequently, it has become common to describe AI as a technology that “creates more time” for humans.

Indeed, generative AI can significantly reduce the amount of time required for many cognitive tasks. It accelerates information search, assists in content creation and analysis, and enables the exploration of multiple perspectives in reasoning processes. In this sense, AI represents a transformative technology that fundamentally changes how humans utilize their limited time.

However, from the perspective of the time resource theory developed in this study, the expression that “AI increases time” requires careful reconsideration. AI does not create time itself. The total amount of time available to human beings remains unchanged; a day still consists of twenty-four hours.

More importantly, AI itself is not the subject of time resources. AI does not possess its own future as a field of personal possibility, does not establish intrinsic goals, and does not exist as a finite living being that experiences the limitation of time. Therefore, AI cannot manage or allocate time as a resource for itself.

This point is crucial because, according to the theoretical framework proposed in this study, time becomes a resource only when a subject can utilize future possibilities in accordance with its purposes. A resource is not defined by the mere existence of an object or quantity, but by the relationship between a subject, its goals, and the possibilities available for action.

From this perspective, what AI directly expands is not time itself, but rather **Choice-Possible Time**—the portion of future time in which individuals can exercise meaningful choices.

For example, suppose a task that previously required one hour of information gathering can now be completed within a few minutes through generative AI. The saved time can then be reallocated toward research, creative activities, learning, family interaction, or rest. What has increased in this process is not the total quantity of time available to the individual. Rather, what has expanded is the range of possible actions that the individual can choose within their remaining time.

Therefore, the essential function of AI is not the replacement or expansion of human time itself. Instead, AI transforms the structure of human time utilization by expanding

the range of choices available within future time. It enables individuals to allocate their limited time resources more flexibly according to their purposes and values.

This interpretation is also consistent with the author's previous argument that large language models (LLMs) do not possess values of their own but provide a **Reasoning Space** in which human reasoning can be expanded and reconstructed. AI is not an autonomous agent that determines human values or purposes. Rather, it functions as an intellectual infrastructure that enables humans to reconsider their values, generate new questions, and reorganize the allocation of their time resources.

In this sense, the fundamental significance of AI in the coming era does not lie in increasing the amount of time itself. Instead, the essence of AI lies in expanding human **Choice-Possible Time**—the capacity of individuals to transform limited temporal resources into meaningful possibilities.

Chapter 8

Conclusion

This study has critically examined the widely accepted assumption that “time is a resource” by reconsidering the fundamental concept of resources itself. In contemporary society, time is routinely treated as a resource through practices such as time management, productivity improvement, and scheduling. However, when examined more carefully, time itself does not possess the fundamental characteristics traditionally associated with resources. Time cannot be owned, stored, accumulated, or transferred to others. Therefore, there exists a conceptual difficulty in treating time itself as an intrinsic resource in the same sense as conventional material resources.

This study argues that this difficulty arises because previous discussions have often failed to distinguish between **time itself** and **the resource potential contained within time**. To address this problem, this research reconstructed the concept of resources from a relational perspective. Rather than defining resources by their material properties, this study proposed that resources emerge when something becomes available for a subject to utilize in pursuit of a goal.

From this perspective, time itself is not a resource. Instead, time becomes a resource only when a subject can utilize a given temporal interval as a means of achieving its purposes. Thus, the relevant object of analysis is not abstract, flowing time, but the time intervals that become available for human action.

Furthermore, this study demonstrated that the existence of a time interval alone is insufficient for establishing resource value. A period of time may be clearly defined as an interval while still lacking meaningful resource potential. For example, compulsory meetings, restricted travel periods, or other forms of constrained time may constitute recognizable temporal intervals, yet they provide limited opportunities for autonomous action. Therefore, time becomes a resource only when three conditions are simultaneously satisfied: the presence of a subject, the existence of a purpose, and the availability of meaningful choices.

Based on this theoretical framework, this study introduced the concept of **Choice-Possible Time**. Choice-Possible Time refers to time during which a subject can choose among multiple possible actions in accordance with its own purposes and intentions. This concept extends conventional approaches that evaluate time primarily through quantity by incorporating the qualitative dimension of time: the degree to which time enables meaningful choice and action.

Through this concept, the study also reconstructed the relationship between past, present, and future. Future time represents a domain in which possibilities remain open and choices have not yet been determined. Present time is the process through which these possibilities are transformed into actual actions. Past time represents the accumulated history of previously consumed choices. From this perspective, human life is not merely the passage of chronological time; rather, it is the historical process of how individuals have allocated and utilized their Choice-Possible Time.

The implications of this framework extend to the emerging relationship between humans and artificial intelligence. This study argued that generative AI does not create additional time itself. The total amount of human time remains unchanged. Instead, AI expands human Choice-Possible Time by reducing the effort required for certain cognitive tasks and enabling individuals to redirect saved time toward alternative purposes, such as learning, creativity, research, relationships, or rest.

AI itself is not a subject of time resources because it does not possess intrinsic purposes, personal futures, or finite human existence. Nevertheless, it functions as an intellectual infrastructure that expands the range of possibilities available to human beings. In the AI era, the significance of technology should therefore not be understood as increasing the quantity of time, but as enhancing the possibility space within human time.

The central proposition of this study can therefore be summarized as follows:

Time is not inherently a resource. Time becomes a resource when a subject, guided by purposes, can exercise meaningful choices within it.

The concept of **Choice-Possible Time** provides a theoretical framework for evaluating this process of resource formation.

This perspective contributes not only to the reconsideration of time but also to the broader philosophical and economic question of what constitutes a resource. Resources are not simply objects that exist independently in the world. Rather, they emerge through the relationship between subjects, purposes, and available possibilities for action. By reconstructing the concept of resources through the analysis of time, this study provides a new theoretical perspective for the fields of time economics, resource theory, and the understanding of human agency in the age of artificial intelligence.

Ultimately:

Time is not a resource by itself.

Time becomes a resource only when human beings possess the ability to choose their future.

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